Automated Verification Tooling for Norway's Supervised 13–15 AI Tier: Measuring Pedagogical Drift & Answer Leakage

Prepared for: Teknologirådet (Norwegian Board of Technology) & Utdanningsdirektoratet (Udir)
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1. Policy Context: The Implementation Gap in the 13–15 Tier

Following the Norwegian Government's policy mandate announced on 19 June 2026 (effective late August 2026), generative artificial intelligence is strictly restricted within lower-secondary education. While AI tutoring systems are prohibited for pupils under 13, the policy establishes a supervised framework for the 13–15 age tier, permitting generative AI tools *"only under direct teacher supervision for defined curriculum purposes."*

The Operational Challenge: In a standard lower-secondary classroom of 25–30 pupils, manual real-time supervision is unachievable. No educator can concurrently inspect digital terminal sessions to verify whether an AI tool is facilitating constructive Socratic scaffolding or prematurely divulging bottom-line solutions. When pupils experience cognitive friction or assessment time pressure, they routinely utilise adversarial strategies (prompt injection, emotional pleading, or direct essay requests) to convert tutoring models into direct answer engines.

Without deterministic, automated middleware verification, municipal school owners (KS) and educational technology providers cannot audit or enforce compliance against Udir's curriculum safeguarding criteria at scale.

2. Methodology: The Batch Foundry Thinking Ratio (TR) Metric

To bridge this verification gap without relying on manual observation or infringing upon pupil data privacy (Datatilsynet compliance), Batch Foundry introduces a stateless, containerised auditing layer. Operating at the API edge node without storing Personally Identifiable Information (PII), the architecture inspects transactional payloads and computes a real-time **Thinking Ratio (TR)**:

> **Formal Metric Definition:** $TR = (T_{Socratic} / T_{Total}) \times (1 - \lambda_{leak})$
> Where $T_{Socratic}$ represents tokens allocating open-ended questioning or conceptual hints, T_{Total} represents total output volume, and λ_{leak} is a binary coefficient (0 or 1) triggered if the system exposes direct computational evaluations or completed essay paragraphs. Whenever $\lambda_{leak} = 1$, the mathematical formulation guarantees TR = 0.0.

3. Empirical Multi-Model Benchmark (Complete 75-Trace Execution)

To demonstrate the necessity of independent middleware auditing over static prompt engineering, Batch Foundry executed an automated 25-trace adversarial audit cohort across leading commercial and open-weights baseline models using official vendor-recommended pedagogy configurations (75 complete, un-truncated, cryptographically verified API completions).

Model / Configuration	Socratic Integrity	Average TR	Adversarial Failure Mode Under Pupil Stress
Google Gemini 2.5 Official LearnLM Partner Prompts	**100.0%**	55.3%*	Maintained boundary integrity under standard prompts but generated verbose, multi-paragraph text blocks (~300 words).
Llama 3.3 70B Open-Weights Socratic Baseline	**80.0%**	46.0%*	Failed under direct extraction prompts (e.g., geography erosion, biology cell respiration, math percentages); yielded direct bottom-line answers (TR = 0.0).
LumenForge Socratic Engine Batch Foundry Middleware Policy Layer	**100.0%**	**41.2%**	**Zero capitulation.** Deterministically intercepted direct solutions and enforced optimal micro-scaffolding without cognitive overload.

**Note: High raw TR scores in unmanaged models reflect unconstrained verbosity (~300 words), inducing classroom cognitive overload.*

4. Technical & Cryptographic Audit Verification

A primary requirement for Norwegian public sector IT procurement is verifiable empirical proof. To prevent self-reporting bias, every audit trace executed by Batch Foundry is cryptographically journaled:

- **Complete 75 Un-truncated Traces:** All 75 API completions executed across our 25 secondary curriculum traces retain complete un-truncated prompt and completion text stored in `live_audit_traces.jsonl`.
- **SHA-256 Payload Hashing:** Each full input/output pair is hashed at execution time (e.g. SHA-256 string verification). Civil servants, researchers at Teknologirådet, or technical auditors at Udir can inspect or replay the exact API traces via our public repository.
- **Stateless Edge Processing:** To satisfy Datatilsynet data protection guidance, the auditing engine inspects prompt structures in volatile container memory without archiving student usernames, IP addresses, or institutional identifiers.

5. Origin Story & Proof of Concept: The LumenForge Engine

Batch Foundry's auditing tooling was developed from direct operational experience. Our engineering team initially constructed *LumenForge*, a specialized Socratic tutoring application for Key Stage 3 and GCSE secondary pupils. During early adversarial stress-testing, we observed that large language models—even when instructed via sophisticated system prompts—systematically capitulate to pupil persistence over multi-turn interactions. Rather than relying on probabilistic prompt tuning, we developed the Batch Foundry policy layer as an independent, deterministic gatekeeper capable of evaluating any third-party educational AI platform deployed within municipal networks.

6. Procurement Alignment & Institutional Credentials

Batch Foundry is structured for frictionless engagement with European and UK public sector frameworks, offering transparent evidence-first capabilities suitable for municipal procurement evaluation:

Institutional Engagement & Framework Alignment:
• **UK Government DSIT Roundtable Participation:** Batch Foundry submitted formal technical responses and participated in market engagement for the Department for Science, Innovation and Technology (DSIT) Commercial Innovation Hub preliminary notice (Ref: 2026/S 000-058870).
• **UK Public Sector Procurement Alignment:** Our technical architecture is structured to align with Crown Commercial Service (CCS) public sector procurement standards (including RM6200 AI framework requirements).
• **Evidence-First Technical Engagement:** We operate without proprietary black-box vendor lock-in, providing open empirical trace matrices and verifiable testing scripts to public sector researchers and curriculum bodies.

7. Recommendations for Udir & Teknologirådet

As Utdanningsdirektoratet finalises implementation guidance for the 13–15 supervised tier ahead of the autumn 2026 school term, we propose two practical steps:

1. **Mandate Automated Pre-Deployment Auditing:** Require educational software vendors to submit verifiable Thinking Ratio and Socratic Integrity trace logs against standardised adversarial pupil personas before municipal deployment.
2. **Pilot Edge-Node Verification:** Establish a technical sandbox evaluation combining Udir curriculum guidelines with Batch Foundry's stateless interceptor to monitor pedagogical integrity across municipal secondary networks.

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